HROC · NCAN · Deep-Learning Track

Does the cortex change as the spinal cord learns?

First real-data results — Animal 9 through the full pipeline, end to end.

Suchith Rao • prepared for Dr. Carp • July 2026

THE SCIENTIFIC BET

The question, in one line

The paradigm

1 **Baseline**

Record the spinal H-reflex (a leg-muscle reflex) without training.

2 **Down-conditioning**

Reward the animal for making the reflex smaller. The spinal cord slowly learns.

3 **The brain?**

We record cortex (ECoG) at the same time — and ask if IT changes too.

What's new here

Chad already published the simple link — when the reflex goes down, ECoG activity shifts. That correlation is done.

Our new question:

Does the brain's internal representation drift across conditioning — a cortical fingerprint of learning that's never been shown?

THIS WEEK

From a synthetic demo to real data

BEFORE

The pipeline ran on synthetic (fake) data shaped like Animal 9 — enough to prove the machine works, but not a scientific result.

NOW

The exact same pipeline is trained on the real Animal 9 recordings — decoded, phase-labeled, and run end to end.

283,315

real trials decoded

3

conditioning phases recovered

2

brain + muscle channels

ORIENTATION

What one trial actually is

Animal 9's recordings live in a 2006 lab database. Each stimulus creates one trial — a 30-millisecond snapshot, sampled 5,000 times a second, on two electrodes at once:

A Muscle — EMG (soleus)

The leg-muscle response to the stimulus. This is where the **H-reflex** lives — the thing conditioning trains.

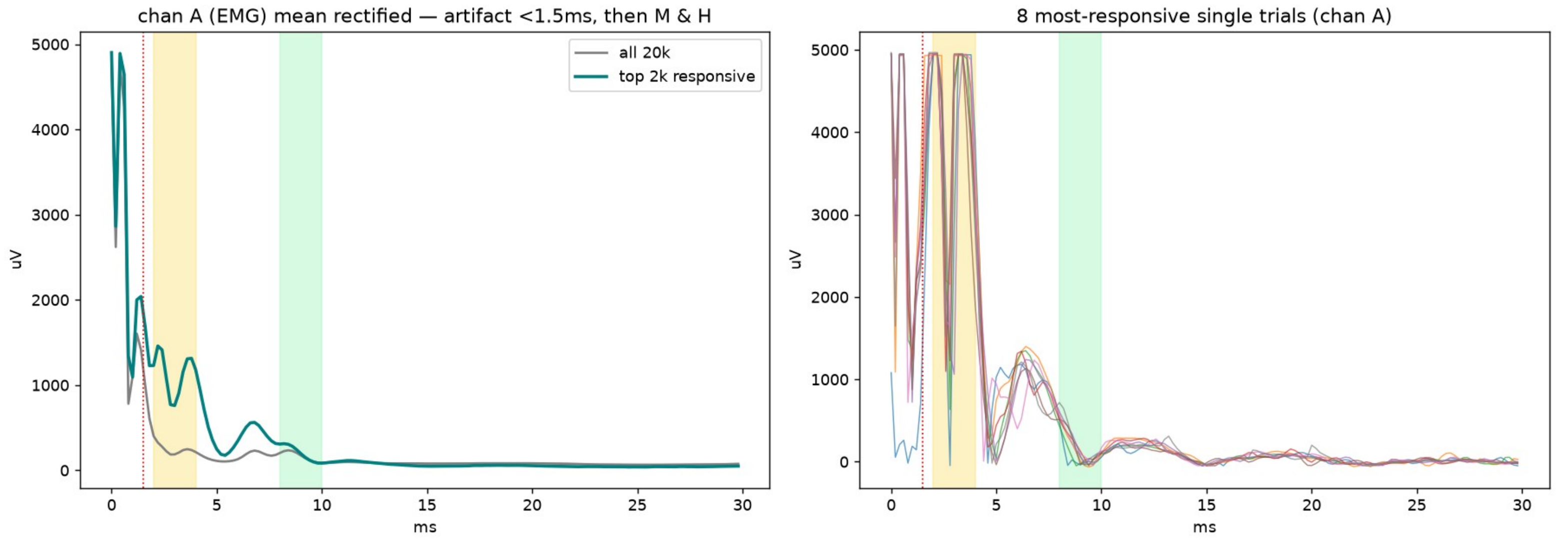
→ our prediction target

B Brain — ECoG (cortex surface)

Electrical activity from the surface of the motor cortex, recorded at the very same moment as the muscle. This is the brain signal we study.

→ the model's input

First: prove we're reading the signal right



What this took

- Loaded the 2.3 GB database locally
- Cracked the byte format (it was mislabeled in the lab docs)
- Confirmed channel A = muscle, B = brain
- Verified against textbook M / H timing

The raw data is a 20-year-old binary format. We confirmed the decode by averaging the muscle channel: it shows the textbook shape — a stimulus artifact, then the M-wave (2–4 ms), then the H-reflex (~6–7 ms). Real physiology, not noise.

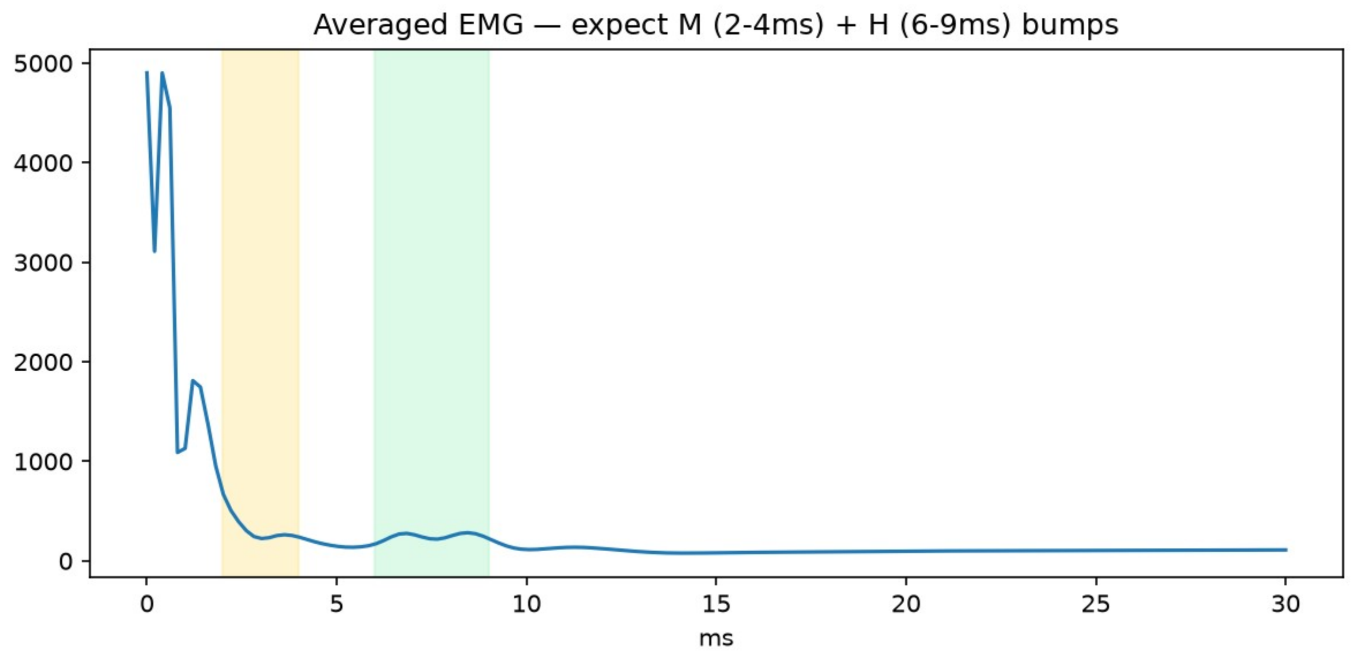
Recover the conditioning phases

We read the experimenter's own typed log from 2006 to find when each phase began — then tagged all 283k trials by time.

Baseline	Down-conditioning	Post
149,064	**55,397**	**78,854**
Setup + characterize the reflex	Cord trained to shrink the reflex	After conditioning stopped

Key detail: Animal 9 ran Baseline → Down-conditioning → Post — NOT the full 6-phase protocol. We verified the phase boundaries against the authoritative database schema, so these labels are trustworthy.

Define what we predict: the H-reflex size



The measure

Where: 6–9 ms — the experimenter's log says exactly that, and it matches what we measured.

How: mean rectified amplitude in that window — your spec, not a single fragile peak.

Green band = the H-reflex window on the real grand-average.

THE MODEL, IN PLAIN WORDS

A 'cortical fingerprint' of each trial

Instead of hand-picking features, the model learns its own compact summary of the brain signal — call it a fingerprint (48 numbers per trial). It's trained in two stages:

1 Learn the fingerprint

The model compresses each brain trace down to 48 numbers, then rebuilds the original from them. To succeed, those numbers must capture the real structure of the signal.

No labels used here — it never sees the H-reflex.

2 Test what it captured

We freeze the fingerprint so it can't change, then attach a small predictor and ask: can you read the leg's H-reflex size from the brain fingerprint alone?

If yes → the brain signal genuinely carries H-reflex information.

PLAIN ENGLISH — A QUICK CHEAT SHEET

How to read the next three slides

The fingerprint

48 numbers the model invents to summarize each brain trace. Nobody tells it what to look for — it finds its own summary.

R^2 (a 0-to-1 score)

How much of the reflex's trial-to-trial change the fingerprint can explain. 0 = nothing, 1 = perfect. We get 0.17 — weak but real, above chance.

Drift

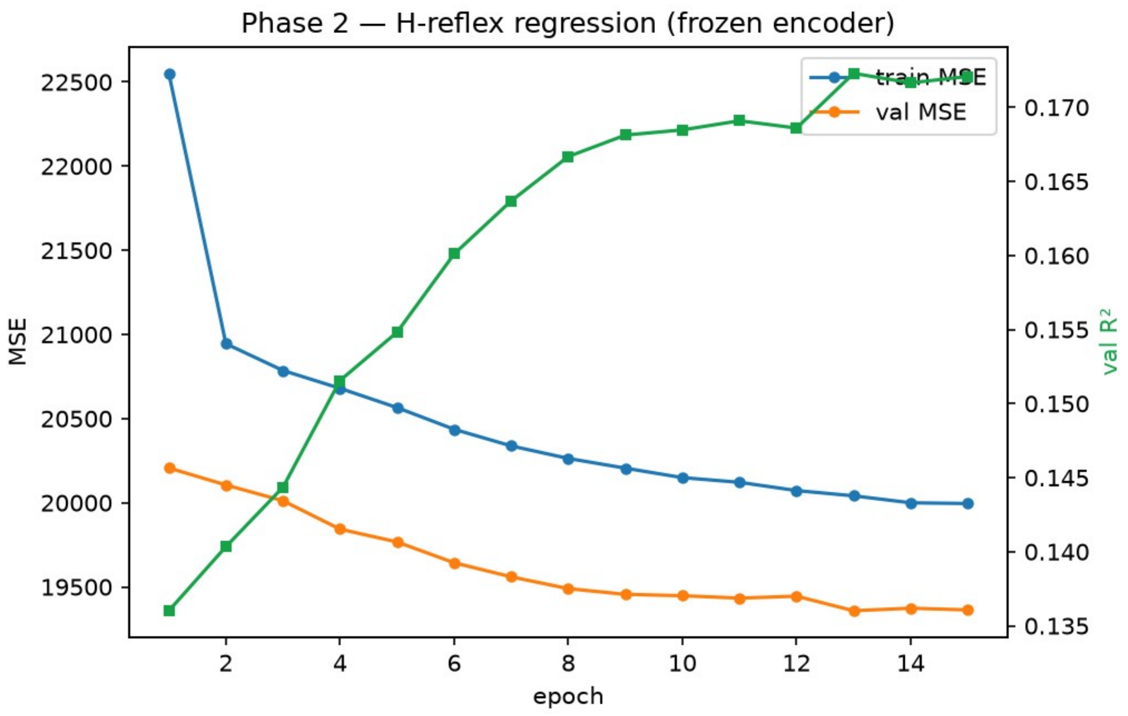
How far the average brain state has moved from where it started at Baseline. A bigger number means the representation changed more.

The catch

Some of that movement might come from the recording setup changing, not from learning. That's the one thing we still have to rule out.

RESULT 1

The fingerprint really does carry the reflex



$R^2 = 0.17$

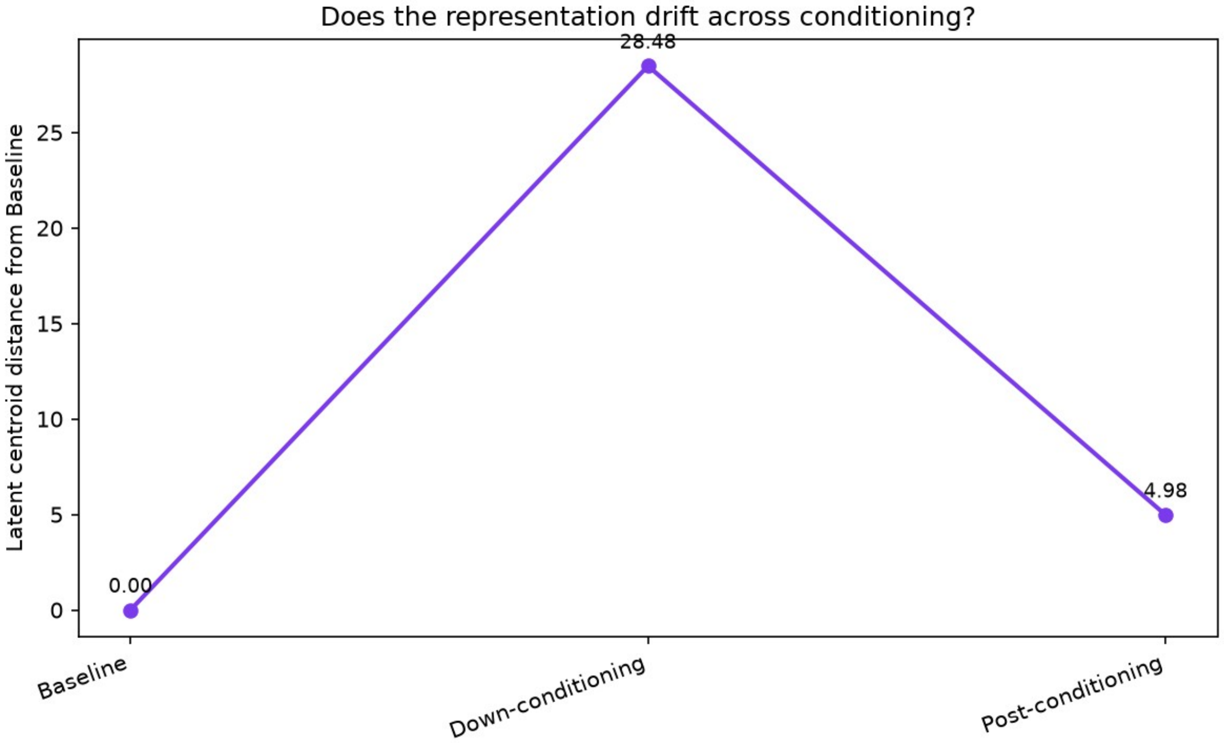
brain fingerprint → H-reflex, frozen encoder

Read it honestly. The cortical fingerprint predicts ~17% of the reflex's variation — with no labels during learning. Modest, but real and well above chance.

On synthetic data this was 0.90 — only because that data was built to encode the reflex. 0.17 is the real cortex.

RESULT 2 · THE HEADLINE

The cortical fingerprint drifts as the cord learns



What you're seeing

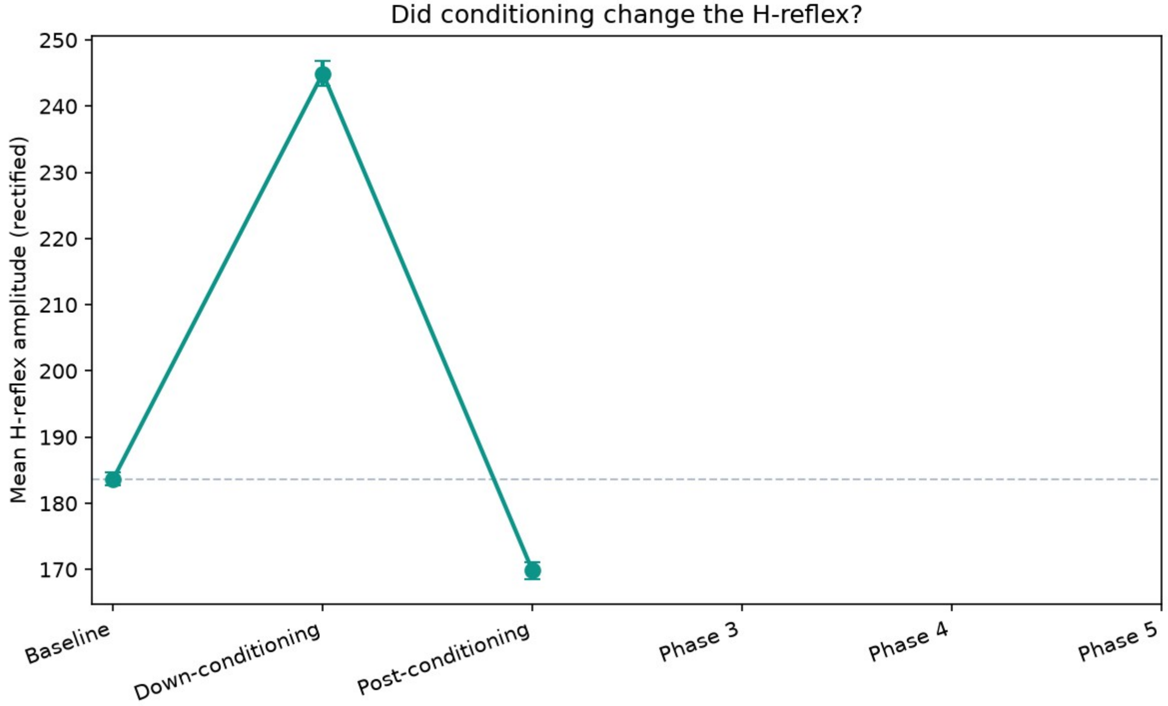
Each point = how far that phase's average brain fingerprint sits from Baseline.

It jumps to 28.5 during down-conditioning, then falls back to 5.0 once conditioning stops.

The representation moves while the cord is learning — and partly recovers after. That's the effect we came to find.

THE HONEST CAVEAT — READ THIS

Is the drift learning, or just session state?



The problem

Down-conditioning should make the H-reflex SMALLER. But in the data it's larger during down-conditioning than at Baseline.

Most likely why:

our 'Baseline' is really the setup period — the log shows the stimulus being constantly re-tuned, so responses there are smaller and inconsistent.

So part of the cortical drift may track session state (fixed-stimulus, active recording) rather than learning itself. #1 thing to resolve.

WHERE WE STAND

Is this good progress? My honest scorecard

STRONG

Infrastructure

Real data flows end to end — decode, phase labels, training, analysis. Reproducible, on a laptop. This was the hard part and it's done.

PROMISING

First real signal

The fingerprint predicts the reflex above chance (R^2=0.17) and the representation clearly moves during conditioning.

NOT YET

A publishable claim

Needs the baseline confound resolved, proper statistics, and more animals. Today is a strong foundation, not a result.

NEXT

The plan — and it scales

Animal 9 is 1 of ~14 conditioned animals sitting in the Drive. The pipeline is built to pool them — every new animal makes the drift result stronger.

1	**Resolve the baseline confound**	Restrict 'Baseline' to post-setup trials; regress out raw reflex size. Does the drift survive?
2	**Add statistics**	Bootstrap the drift — a within-baseline split should drift ~0. Is 28.5 vs 5.0 real?
3	**Pool more animals**	Convert 6, 10, 11, 12, 13... one at a time, watching for animal-identity confounds.
4	**Bring in controls**	Up-conditioning animals should drift the other way; freely-running should barely drift.

WHERE I NEED YOUR READ

Four neuro calls — not code

1 Baseline definition

Given the setup-period confound, how should we define a clean behavioral baseline for Animal 9?

2 The H-reflex label

Confirm: mean rectified amplitude, 6–9 ms window — is that the measure you want us predicting?

3 What would convince you

What controls make the cortical drift believable — up-conditioning reversing it? freely-running as a null?

4 Which animals next

Of the ~14 in the Drive, which should we prioritize converting and pooling?

Code + all figures on GitHub — happy to add you and Tarun as collaborators.